



An Internship Report
Of
Data Analyst

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IILM University, Greater Noida, U.P

In partial fulfilment of the requirement of degree of
B.Tech (CSE)

BY

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Name Of the Student :- Anshika Singh

Batch :- 2026-27

CANDIDATE'S DECLARATION

I, ANSHIKA SINGH, do hereby declare that the internship report titled "Internship Report on Data Analytics" has been completed by me to fulfil the requirement for the award of the degree of Bachelor of Technology in Computer Science Engineering. All the references have been duly acknowledged, and I have not incorporated any material from any other source without citation. I affirm that this report is entirely my own work and has not been submitted, in full or in part, to any other institute for the award of any degree or diploma, and I shall be solely responsible for any copyright violation arising in this regard.

Signature

Student Name: ANSHIKA SINGH
Roll No.: 25SCS1003004925

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to BeeSkilled for giving me the opportunity to undertake this six-week internship in the Data Analyst domain. This internship exposed me to practical, real-world data analysis practices and allowed me to apply the concepts learned during my academic curriculum in a hands-on setting.

I am thankful to IILM University, Greater Noida, U.P. and the School of Computer Science and Engineering for providing the academic foundation and encouragement that made this internship possible. I extend my heartfelt thanks to my faculty members and academic support staff for their guidance, constructive feedback, and continued support throughout this internship period.

I am also grateful to everyone at BeeSkilled who supported me during this internship for sharing their knowledge and insights on various aspects of Data Analytics. Finally, I express my deepest gratitude to my family for their constant encouragement and support throughout this journey.

Signature

Student Name: ANSHIKA SINGH
Roll No.: 25SCS1003004925

The certificate below was issued by BeeSkilled on 28 August 2026, confirming successful completion of the 6-week online internship program in the Data Analyst domain.



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CHAPTER 4: PROJECT DESCRIPTION

4.1 Introduction

Data Analytics involves converting raw and unprocessed information into meaningful insights that can assist in understanding trends, evaluating performance, and making better decisions. With businesses increasingly relying on data for their day-to-day operations, the ability to examine and interpret information systematically has become an important part of modern decision-making. The field combines several approaches, including spreadsheet analysis, programming-based data processing, statistical techniques, and interactive business intelligence tools.

A reliable analysis begins with preparing the underlying data. Datasets obtained from practical sources may contain incomplete entries, repeated records, inconsistent formats, or values stored in unsuitable formats. Identifying and resolving such issues is therefore an essential part of the analytical process, as the quality of the input data directly influences the accuracy of the results. After preparation, the data can be examined and transformed using tools such as Microsoft Excel, Python and Pandas, and Power BI. These technologies make it possible to perform calculations, explore datasets, generate business metrics, and present analytical results through visual dashboards.

This report presents the work completed during a six-week Data Analyst internship with BeeSkilled, with the practical work documented across four major learning modules. The internship began with Excel-based sales analysis, followed by dataset inspection and statistical analysis using Python and Pandas. The third module focused on data cleaning and transformation, while the final module involved an end-to-end business analytics project incorporating data validation, exploratory analysis, KPI calculation, Power BI measures, and business recommendations. Overall, the internship provided practical exposure to the process of taking raw data through different stages of preparation and analysis and ultimately converting it into useful, decision-oriented information.

4.2 Organization Profile

BeeSkilled is an online platform focused on internships and skill development, operating with the tagline “Tech and Internships.” It provides structured, project-oriented learning programs intended to strengthen students’ practical abilities and provide exposure to industry-relevant tools and workflows. The organization is registered with AICTE (All India Council for Technical Education) and MSME (Ministry of Micro, Small and Medium Enterprises), as stated in the internship-related communication.

The internship covered in this report was completed in the Data Analyst domain through a structured six-week online internship program. The program combined weekly learning modules with practical assignments and an end-to-end capstone project, allowing the concepts covered during training to be applied to datasets and analytical tasks. The program was conducted under the guidance of Ms. Simranjeet Kaur, Founder & Program Head, with the internship completion certificate awarded after the required coursework and capstone activities were successfully completed.

4.3 Problem Statement

In the present business environment, organizations continuously produce large amounts of data through their daily activities. However, raw data alone does not provide meaningful information unless it is properly prepared, examined, and interpreted. The primary challenge addressed during this internship was to understand how unprocessed datasets can be systematically converted into reliable and useful business insights through a structured data analytics process.

**Raw Data → Data Cleaning → Data Transformation → DataAnalysis → Visualization →
Insights → Recommendations**

This process was implemented through different practical activities during the internship. In Excel, a large sales dataset was summarized using Pivot Tables to obtain meaningful business metrics. Python and Pandas were used to inspect a transactional dataset and perform data-cleaning operations on an exercise-tracking dataset containing missing and duplicate records. During the Week 4 capstone project, a sales transaction dataset was taken through data cleaning, validation, exploratory analysis, KPI evaluation, and business interpretation to develop meaningful findings and stakeholder-oriented recommendations.

4.4 Project Objectives

- To develop a fundamental understanding of Data Analytics and recognize its importance in supporting business decisions.
- To gain practical experience in Excel-based analysis, including the use of Pivot Tables and aggregation techniques for large transactional datasets.
- To examine sales, revenue, and category-level performance through Pivot Table-based summaries and dashboards.
- To strengthen Python fundamentals and apply the Pandas library for systematic exploration of structured datasets.

- To use essential Pandas functions, such as head(), info(), shape, and describe(), for understanding and summarizing datasets.
- To perform data-cleaning operations by detecting and appropriately handling missing values and duplicate records.
- To process inconsistent date information by converting date fields into suitable datetime formats and applying appropriate methods for missing-value treatment.
- To generate new analytical variables by deriving calculated columns from existing dataset attributes.
- To extract relevant information from datasets by applying filters based on defined analytical or business conditions.
- To gain an understanding of Power BI dashboard development and the use of DAX for creating calculated business measures.
- To develop data storytelling skills by presenting analytical findings in a clear and understandable manner for stakeholders.

4.5 Scope of the Project

The internship focused on structured, dataset-oriented data analysis using sample and practice datasets provided through the BeeSkilled curriculum. The work was carried out in a learning environment and did not involve the use of live production systems or confidential organizational data. The major areas covered within the scope of the internship were:

- Working with structured business datasets, particularly sales and order-related data, and organizing them for analysis.
- Performing spreadsheet-based analysis using Microsoft Excel, including Pivot Tables, aggregation, and summarization of important business metrics.
- Dataset inspection and exploration using Python and the Pandas library.
- Data cleaning and transformation, including missing-value handling, duplicate removal, and date-format correction.
- Preparing datasets for analysis by identifying missing values
- Transforming datasets for analytical purposes

- Developing business intelligence measures using DAX.
- Applying data storytelling techniques .

4.6 Technologies and Tools Used

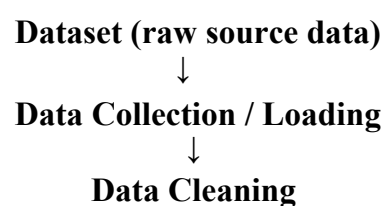
Table 4.1 summarizes the primary tools and technologies used during the internship, along with their purpose and how each was applied.

Tool	Purpose	Application during Internship
Microsoft Excel	Spreadsheet-based data organization and analysis	Built a pivot-table dashboard analyzing total revenue, sales by category, yearly trends, and segment-wise revenue from a 51,290-record orders dataset
Python	General-purpose programming for data analysis tasks	Used for dataset loading, inspection, and data-cleaning scripts across Weeks 2 and 3
Pandas	Data manipulation — reading, inspecting, cleaning, and transforming data	Inspected a 200-row sales dataset using <code>head()/info()/describe()</code> ; cleaned a 32-row exercise dataset by handling missing values, duplicates, and date formatting; created a derived column and filtered records
Power BI	Business intelligence dashboarding and calculated (DAX) measures	Built DAX measures — Total Sales, Total Profit, Total Units Sold, Profit Margin, Average Order Value, and Loss-Making Orders — for the Week 4 capstone dashboard

Table 4.1: Technologies and Tools Used

4.7 Data Analytics Workflow

Rather than a software-engineering system architecture, the internship followed a Data Analytics workflow that moves from raw data collection through cleaning, transformation, and analysis, to visualization and, finally, actionable recommendations. This workflow, shown in Figure 4.1, was applied consistently across all four weeks of the internship.



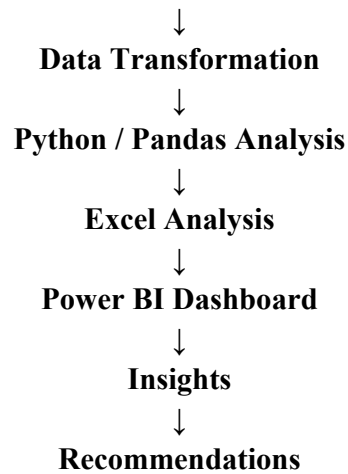


Figure 4.1: Data Analytics Workflow

4.8 Methodology

This section documents the methodology followed during each week of the internship, along with the specific work performed and the results obtained.

4.8.1 Week 1: Excel Data Analysis

The first week of the internship introduced the fundamentals of using Microsoft Excel for data analysis, with emphasis on spreadsheet operations, aggregation formulas, Pivot Tables, and techniques for organizing and summarizing data.

As part of the practical assignment, the Global Superstore Orders dataset was analyzed through Pivot Table-based sales summaries. The dataset consisted of approximately 51,290 records and 24 columns. The category-wise analysis presented in Figure 4.2 indicated that Technology generated the highest sales of approximately \$4.74 million, followed by Furniture with around \$4.11 million and Office Supplies with approximately \$3.79 million.

The year-wise analysis also indicated a steady increase in sales during the observed period, with revenue growing from approximately \$2.26 million in 2012 to approximately \$4.30 million in 2015. This exercise provided practical experience in using Excel Pivot Tables to convert a large transactional dataset into concise and interpretable business summaries.

<i>Sales by Category</i>	<i>Total Sales</i>
Technology	\$4.74M
Furniture	\$4.11M
Office Supplies	\$3.79M

<i>Yearly Sales Trend</i>	<i>Total Sales</i>
2012	\$2.26M

<i>Yearly Sales Trend</i>	<i>Total Sales</i>
2013	\$2.68M
2014	\$3.41M
2015	\$4.30M

Figure 4.2: Excel Pivot Table — Sales Performance Analysis (Category-wise and Yearly)

Customer-segment-wise revenue was also analyzed using pivot tables, shown in Figure 4.3. The Consumer segment generated the highest revenue at approximately \$6.51M, followed by Corporate at approximately \$3.82M and Home Office at approximately \$2.31M.

<i>Customer Segment</i>	<i>Total Sales</i>
Consumer	\$6.51M
Corporate	\$3.82M
Home Office	\$2.31M

Figure 4.3: Segment-wise Revenue Analysis

This exercise demonstrated how Excel pivot tables can be used to organize a large raw transactional dataset into summarized, decision-ready business metrics using SUM-based aggregation, without writing any code.

4.8.2 Week 2: Python/Pandas Data Analysis

The second week of the internship focused on Python and the Pandas library for structured dataset analysis. The topics included importing datasets with Pandas, examining their structure and data types, and obtaining descriptive statistical summaries. The practical exercise required loading a sales dataset and conducting an initial inspection to understand its contents.

A sales dataset containing 200 records and 10 columns was imported using Pandas. The columns included `order_id`, `customer_name`, `order_date`, `category`, `sub_category`, `product_name`, `quantity`, `unit_price`, `total_price`, and `region`. The dataset was examined using `df.shape` and `df.info()` to verify its dimensions, column structure, and data types, as presented in Figure 4.4.

```
>>> df.shape
(200, 10)

>>> df.info()
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 200 entries, 0 to 199
```

```
Data columns (total 10 columns):
#      Column      Non-Null Count  Dtype
---  -
0  order_id      200 non-null    int64
1  customer_name 200 non-null    object
2  order_date     200 non-null    object
3  category       200 non-null    object
4  sub_category   200 non-null    object
5  product_name   200 non-null    object
6  quantity       200 non-null    int64
7  unit_price     200 non-null    float64
8  total_price    200 non-null    float64
9  region         200 non-null    object
```

Figure 4.4: Dataset Structure Summary (df.shape and df.info() Output)

Figure 4.5 shows the first five records of the dataset retrieved using the `df.head()` method, giving an initial view of the raw transactional data prior to analysis, along with the list of column names retrieved using `df.columns`.

```
>>> df.columns.tolist()
['order_id', 'customer_name', 'order_date', 'category',
 'sub_category', 'product_name', 'quantity', 'unit_price',
 'total_price', 'region']

>>> df.head()
   order_id customer_name order_date      category  quantity
0      1000    <customer>    <date>    <category>         ..
1      1001    <customer>    <date>    <category>         ..
2      1002    <customer>    <date>    <category>         ..
3      1003    <customer>    <date>    <category>         ..
4      1004    <customer>    <date>    <category>         ..
```

Figure 4.5: Sample Dataset Records (df.head() and df.columns Output)

Figure 4.6 shows descriptive statistics generated using the `df.describe()` method for the numeric columns — `quantity`, `unit_price`, and `total_price` — giving a quick statistical overview of the dataset's central tendency and spread. The mean quantity per order was 4.91, the mean unit price was 2421.495, and the mean total price was 12,101.535, with a maximum total price of 47,940 recorded in the dataset.

```
>>> df.describe()
      quantity  unit_price  total_price
count  200.000000    200.000000    200.000000
mean     4.910000    2421.495000   12101.535000
std      ...           ...           ...
min      ...           ...           ...
max      ...           ...    47940.000000
```

Figure 4.6: Statistical Summary of the Dataset (df.describe() Output)

The exercise showed that using `shape`, `info()`, `head()`, and `describe()` together offers an efficient way to understand an unfamiliar dataset at the initial stage. These functions help verify the dataset size, identify column data types, preview sample records, and examine the distribution of numerical values before proceeding with detailed analysis.

4.8.3 Week 3: Python and Data Wrangling

The third week concentrated on data wrangling with Python and Pandas, with emphasis on detecting and correcting common data-quality problems such as missing entries, duplicate records, and inconsistent date values. The practical assignment required cleaning a dataset using Pandas and generating an updated, cleaned version for further use.

For the practical exercise, a workout/activity dataset containing 32 rows and 5 columns—Duration, Date, Pulse, Maxpulse, and Calories—was imported and examined. The initial inspection identified one duplicate record, one missing value in the Date column, and two missing values in the Calories column, as presented in Figure 4.7.

```
Original dataset      : 32 rows x 5 columns
Columns              : Duration, Date, Pulse, Maxpulse, Calories

Data quality issues found:
- Missing Date values      : 1
- Missing Calories values : 2
- Duplicate rows           : 1

Cleaning actions performed:
- Removed 1 duplicate row
- Converted Date column to proper datetime format
- Filled missing Date value using interpolation
- Filled missing Calories values using the column mean
- Added derived column: Calories_Per_Minute
  Calories_Per_Minute = Calories / Duration

Cleaned dataset      : 31 rows x 6 columns
```

Figure 4.7: Data Cleaning / Transformation Summary

Following the data-cleaning process, the Date column, which was initially stored in an inconsistent text format, was converted into an appropriate datetime format. The single missing Date value was handled through interpolation, while the two missing Calories entries were replaced using the mean of the column. After these steps, the dataset contained no missing values. A new calculated field, Calories_Per_Minute, was then generated by dividing Calories by Duration for each record. After removing the duplicate row and adding the derived column, the resulting dataset contained 31 rows and 6 columns and was saved as cleaned_data.csv.

As an additional analysis step, the cleaned dataset was filtered to retain only those records where Calories exceeded 300, and this filtered dataset was exported separately as filtered_data.csv. The exercise highlighted the importance of proper data preparation, as missing information, duplicate records, and inconsistent formatting can affect the reliability of subsequent calculations and analysis.

It also demonstrated that checking and improving data quality should be completed before applying further analytical operations.

4.8.4 Week 4: Data Visualization and Capstone Project

The fourth week emphasized Power BI dashboard development, including the creation of calculated DAX measures and visualization of important business metrics. It also introduced the concepts of data storytelling, with a focus on presenting analytical results as meaningful insights and actionable recommendations for stakeholders. The week concluded with an End-to-End Business Analytics Capstone Project, through which the complete data analytics workflow was applied to a financial and sales transaction dataset.

The capstone dataset consisted of 700 records and 16 original columns, including fields such as Segment, Country, Product, Discount Band, Units Sold, Gross Sales, Discounts, Sales, COGS, Profit, Date, Month, and Year.

As part of the cleaning and validation process, 53 records were found to have blank values in the Discount Band field. Further examination showed that these records had zero discount values, so they were categorized as “No Discount” instead of being removed. This ensured that valid transaction records were retained. The dataset was also checked for duplicate entries, with no duplicates identified.

The cleaned dataset was then used for exploratory business analysis and Power BI dashboard development, with calculated DAX measures used to summarize key business metrics. The resulting dashboard and analysis are presented in Figure 4.8.

<i>Business KPI</i>	<i>Value</i>
Total Sales	\$118,726,350
Total Profit	\$16,893,702
Total Units Sold	1,125,806
Average Order Value	\$169,609
Profit Margin	14.2%
Total Orders	700

Figure 4.8: Power BI Business Performance Dashboard — Key KPI Measures

The dashboard incorporated several DAX measures, including Total Sales, Total Profit, Total Units Sold, Profit Margin, Average Order Value, and Loss-Making Orders. The major findings obtained through the exploratory analysis and Power BI dashboard were as follows:

- Government emerged as the most profitable segment, generating approximately \$11.39M in profit, whereas Enterprise was the only segment with an overall loss of approximately – \$614,546.
- The USA achieved the highest sales among the analyzed countries, with revenue of approximately \$25.03M.
- Paseo recorded the strongest overall performance in terms of both sales and profit, while Carretera ranked lowest on both measures.
- Out of 700 total orders, 58 orders (approximately 8.3%) resulted in losses, with these transactions mainly falling under the Medium and High discount bands.
- The monthly analysis showed that October and December 2014 were among the highest-performing months, whereas September 2013 recorded the lowest sales during the analyzed period.

Based on the analysis, the following recommendations were developed from a stakeholder decision-making perspective:

- Reassess discount-approval policies, particularly for higher discount levels, due to their observed connection with loss-making transactions.
- Establish minimum profit-margin thresholds for orders involving substantial discounts to reduce the risk of unprofitable sales.
- Examine the reasons behind the Enterprise segment's negative profitability before considering any further expansion of the segment.
- Review the pricing and cost structure of Carretera, as the product showed comparatively weak results in both sales and profit.
- Optimize inventory, staffing, and resource planning around the seasonal increase in demand observed between October and December.
- Analyze the unusually low sales recorded in September 2013 to determine the factors responsible for the decline and identify possible corrective measures.

The capstone project consolidated the key skills acquired throughout the internship, including data cleaning, validation, exploratory analysis, and business intelligence measure development, into a complete analytical process that transformed raw transactional data into meaningful insights and actionable recommendations for stakeholders.

4.9 Expected Outcomes / Learning Outcomes

- Practical proficiency in Excel-based data analysis, including Pivot Tables and summarization of key business metrics from large datasets.
- Working knowledge of Python and the Pandas library for systematic dataset inspection, exploration, and basic analysis.
- Hands-on understanding of data-cleaning techniques, including the identification and treatment of missing values, duplicate records, and inconsistent date formats.
- Ability to generate derived or calculated columns and perform condition-based filtering to extract relevant records from datasets.
- Practical exposure to business-intelligence dashboard development and the creation of calculated DAX measures using Power BI.
- Experience applying an end-to-end analytics workflow to transactional data, covering cleaning, validation, exploratory analysis, KPI evaluation, and reporting.
- Improved capability to interpret analytical results and communicate them through clear, practical, and stakeholder-focused recommendations.

4.10 Certificates of Completion and Other Work Proof

The Internship Completion Certificate issued by BeeSkilled, confirming successful completion of the 6-week Data Analyst internship, is presented in the preliminary pages of this report (page iii). Figure 4.9 below reproduces the original internship offer letter issued at the start of the program as supporting communication proof.



Figure 4.9: BeeSkilled Internship Offer Letter

No additional communication or completion emails beyond the documents shown above were available at the time of preparing this report.

CHAPTER 5: BIBLIOGRAPHY / REFERENCES

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